



Haas School of Business
University of California, Berkeley

Syllabus for MBA/EWMBA 247 Descriptive and Predictive Data Mining

Fall 2026 (2 units)

Course Overview

The importance of data cannot be overemphasized. With the recent explosion in large-scale, high-resolution data, managers are increasingly required to make data-driven decisions, rather than rely on experience and intuition alone. A key step in data-driven decision-making is data mining, which enables businesses to extract useful information and valuable insights from otherwise massive and unorganized data. This hands-on, introductory course teaches you the fundamental principles and practical skills you need to develop data mining models that describe and predict the effects of business decisions in a variety of real-world business applications, including (but not limited to) marketing, finance, and operations. Descriptive models provide insights into “What did happen?” and predictive models, “What could happen?” Topics to be introduced include:

- Exploratory data analysis: frequency distributions, cross-tabulation
- Descriptive data mining: k -means clustering, association rules, text mining
- Predictive data mining: binary classification, logistic regression, k -nearest-neighbors, classification trees, ensemble methods, Responsible AI

Class Meetings

Day	Dates	Time	Room
Wednesday	Aug 26 - Oct 28, 2026 (10 weeks)	6:00pm-9:30pm PT	Chou Hall N300

Instructor Luyi Yang
luyiyang@haas.berkeley.edu
Office: [F557](#) (Email for in-person appointments)

Assistant Ziyi Liu zyliu2023@berkeley.edu

Prerequisite 200S Data and Decisions

Optional Textbook

Selected chapters of [Business Analytics](#) (4th edition) by Camm et al.
Publisher: Cengage Learning

Course Format, Topics, and Schedule

The course consists of lectures, in-class discussions, and workshop sessions that provide hands-on opportunities to apply course concepts, with guidance from the instructional team. This course covers three major topics: Exploratory Data Analysis (as the first step to interpret data, and as the last step to convey data), Descriptive Data Mining (using data to identify relationships, aka *Unsupervised Machine Learning*), and Predictive Data Mining (using data to make a prediction, aka *Supervised Machine Learning*). The week-by-week topics covered are as follows:

Session (Wed)	Topic	Reading	Due Before Class (Tue 10 pm)
Week 1 Aug 26	Exploratory Data Analysis Frequency Distributions, Cross Tabulation Applications: online retail, gender bias in Berkeley graduate admissions, dishonesty in mileage reporting for car insurance	Camm Ch 2, 3.2	
Week 2 Sep 2	Descriptive Data Mining k-Means Clustering Applications: customer segmentation in financial advisory services	Camm Ch 5.1	
Week 3 Sep 9	Clustering application: MRI image segmentation Workshop 1: Airbnb, college football, satellite imagery		
Week 4 Sep 16	Association Rules Applications: mining Instacart grocery transactions, interaction-based movie recommendations	Camm Ch 5.2	Concept Check 1, Workshop Report 1
Week 5 Sep 23	Text Mining Applications: ChatGPT app reviews, Cryptocurrency news headlines Workshop 2: web analytics, Obama's Tweets, content-based movie recommendation	Camm Ch 5.3	Concept Check 2
Week 6 Sep 30	Predictive Data Mining Binary Classification Logistic Regression Applications: predicting Oscar winners and auto loan acceptance	Camm Ch 9.1, 9.2 Camm Ch 9.3	Workshop Report 2
Week 7 Oct 7	k-Nearest-Neighbors Applications: predicting loan default and Spotify hits	Camm Ch 9.4	Concept Check 3
Week 8 Oct 14	Classification Trees (and ensemble methods) Applications: spam detection, predicting parole violation Workshop 3: election prediction, spam detection, handwritten digit recognition	Camm Ch 9.5	Concept Check 4
Week 9 Oct 21	Responsible AI and Course Review Applications: predictive algorithms in criminal justice		Workshop Report 3
Week 10 Oct 28	Online Remote Final Exam		

Software

This course will use R and Jupyter (on datahub) as primary computing tools. R is a free, open-source programming language used for statistical computing and graphics. The Jupyter Notebook is a web-based interactive computing platform. The notebook combines live code, narrative text, and rich, interactive outputs. The address of the course dataHub is

datahub.berkeley.edu

You can log in via your Berkeley email address and CalNet account using a web browser.

Note: The course assumes you have no prior background in either R or Jupyter. You will be provided with clear instructions and concise examples on which to base your work.

If you run into an issue with R or Jupyter, it is likely that someone else has run into the same issue in the past. Therefore, you are welcome to seek help from AI, Google, Stack Overflow, etc. However, we recommend trying it yourself before consulting with AI to make the most of your learning experience.

Performance Evaluation

The course grade breakdown is as follows:

Item	Percentage
Class Participation	20%
4 Individual Concept Checks	20%
3 Group Workshop Assignments	30%
Final Exam	30%

Class Participation: You are expected to attend class, be punctual, have read the required materials before class, ask meaningful questions, answer questions, and participate in discussions. Class attendance is required, and do expect “cold calls.”

Individual Concept Checks (CC): A total of 4 individual concept checks contain multiple-choice and simple calculation problems meant to reinforce your conceptual understanding of the data mining methods taught in class. These concept checks will be auto-graded on bCourses.

Group Workshop Assignments: In group assignments, you will analyze (large-scale) data sets and solve data-driven problems using software tools. You work on group assignments during workshop sessions in Weeks 3, 5, and 8, and hand in your write-up within a week after the day of the workshop. Students work in groups of **3 or 4**. You can choose to form your own group or be randomly assigned to a group.

Conscientious contributions to the group assignments are strictly required. To enforce this policy, I will solicit feedback from each of you about the contributions of your team members to group work. Failure to contribute adequately to the group will result in a grade penalty in this course. In addition, a pattern of such behavior will result in an academic performance review.

Final Exam: You will take an online exam to demonstrate your knowledge of data mining.

Accommodation for Disabled Students Program (DSP) Students

Students with disabilities seeking academic accommodations in this course need to promptly notify the instructor by providing a Letter of Accommodation from the Disabled Students Program.

AI Policy

Students are encouraged to use generative AI tools like ChatGPT in this course, especially for coding assignments. However, you will receive clear instructions on when and where not to use these tools (e.g., during exams). Using these tools outside the provided guidelines is considered a breach of the Berkeley Haas Code of Conduct.

Academic Integrity

UC Berkeley Honor Code: As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others. The Haas School of Business has a zero-tolerance policy for academic dishonesty. Any commission of academic dishonesty, including receiving or giving unauthorized assistance during an exam, will be immediately referred to the campus student conduct committee.

Please review the code of conduct of the [MBA program](#), [EW MBA Program](#), and [UC Berkeley](#) at large.

Instructor Bio

Luyi YANG is an associate professor of Operations and Information Technology Management at the University of California, Berkeley's Haas School of Business. His research interests include service operations, marketplace operations, and sustainable operations. He has studied a broad range of research topics, including the right to repair, food waste, ride-sharing, digital content platforms, social commerce, consumer referral programs, omnichannel retail, and livestream shopping. His award-winning research has been published in leading journals and outlets such as *Management Science*, *Operations Research*, and *Harvard Business Review*, featured by major media outlets such as Forbes, Fox Business, NBC, and Yahoo Finance, and presented at major companies and organizations such as Uber Technologies, the Federal Trade Commission, and IBM Research. He has taught courses in business analytics, data mining, and operations management. Prior to joining Berkeley Haas, he was an assistant professor of operations management and business analytics at Johns Hopkins University's Carey Business School. He received his PhD and MBA from the University of Chicago, Booth School of Business, and his BS in Industrial Engineering and BA in English, both from Tsinghua University.